

Electric Current and DC Circuits, solved

Step-by-step solutions to every practice problem in Chapter 17. Try each problem on your own before you read its solution.

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P17.1

The wires are already filled with free electrons along their whole length. Closing the switch establishes an electric field throughout the circuit almost instantly. The field is set up by surface charges that rearrange at nearly the speed of light. Every free electron in the circuit, including those in the lamp's filament, starts to drift at essentially the same moment. The individual electrons move slowly, but the *signal* that sets them moving, and the energy carried by the fields around the wires, travel at close to the speed of light.

P17.2

Resistances (from $R = V^2/P$ at their rated 230 V):

$$R_{60} = \frac{230^2}{60} = 882 \, \Omega, \quad R_{100} = \frac{230^2}{100} = 529 \, \Omega.$$

In series, the same current flows through both:

$$I = \frac{230}{882 + 529} = 0.163 \, \text{A}.$$

$$P_{60} = I^2 R_{60} = 23 \, \text{W}, \quad P_{100} = I^2 R_{100} = 14 \, \text{W}.$$

The 60 W bulb glows more brightly. With equal currents, the larger resistance dissipates more power. (Both glow much more dimly than normal. In reality their filaments are cooler, so their resistances are lower than calculated, but the conclusion stands.)

P17.3

In parallel:

1. Every appliance receives the full mains voltage it was designed for.
2. Each one can be switched on and off independently, and a fault or a blown bulb in one does not cut off the others.
3. Each draws only the current it needs.

In series, the supply voltage would be shared among all the devices, and it would change whenever one was switched on or off. One open switch or failure would stop everything.

P17.4

The battery's terminal voltage is $V = \mathcal{E} - Ir$. The starter motor draws a very large current, often 100–300 A, so even a small internal resistance, say 0.02Ω , causes a large drop: $Ir = 200 \times 0.02 = 4$ V. The terminal voltage, which also supplies the headlights, falls from about 12.6 V to about 8–9 V, and the lights dim. Cold weather raises the internal resistance further. That is why weak batteries fail to start cars on cold mornings.

P17.5

Same resistance and length. From $R = \rho L/A$, the area must be proportional to the resistivity:

$$\frac{A_{\text{Al}}}{A_{\text{Cu}}} = \frac{2.8}{1.7} = 1.65.$$

The aluminium wire must be 65% thicker in cross-section.

Mass is proportional to density $\times$ area:

$$\frac{m_{\text{Al}}}{m_{\text{Cu}}} = 1.65 \times \frac{2700}{8960} = 0.50.$$

The aluminium conductor weighs **half** as much, and it is also cheaper. For overhead lines, where the towers must carry the weight, this decides the choice. Aluminium strands are usually wound around a steel core for strength.

P17.6**Resistance:**

$$R = \frac{V^2}{P} = \frac{230^2}{1000} = 53 \, \Omega.$$

Length. With $A = \pi(0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7} \, \text{m}^2$:

$$L = \frac{RA}{\rho} = \frac{52.9(1.96 \times 10^{-7})}{1.1 \times 10^{-6}} = 9.4 \, \text{m}.$$

That is why heating elements are tightly coiled. (Nichrome's resistance rises somewhat with temperature, which a precise design would take into account.)

P17.7

- EMF = open-circuit voltage = 9.0 V.
- Internal resistance, from $V = \mathcal{E} - Ir$:

$$r = \frac{\mathcal{E} - V}{I} = \frac{9.0 - 8.0}{0.50} = 2.0 \, \Omega.$$

Check: $I = 9.0/(16 + 2.0) = 0.50 \, \text{A}$. ✓

- Short-circuit current: $\mathcal{E}/r = 4.5 \, \text{A}$.

P17.8**Equivalent resistance.**

$$6 \parallel 12 = \frac{6 \times 12}{18} = 4.0 \, \Omega; \quad 4.0 + 8.0 = 12 \, \Omega; \quad 12 \parallel 12 = 6.0 \, \Omega.$$

Currents.

- Total current: $I = 18/6.0 = 3.0 \, \text{A}$.
- The two $12 \, \Omega$ branches have the same 18 V across them, so each carries 1.5 A.
- The $8.0 \, \Omega$ resistor is in one of these branches, so it carries 1.5 A. Within that branch, the $6 \, \Omega$ and $12 \, \Omega$ resistors share the 1.5 A as 1.0 A and 0.5 A.

P17.9

Single loop. The two EMFs oppose each other, positive to positive:

$$I = \frac{12.6 - 10.0}{0.020 + 0.010 + 0.050} = \frac{2.6}{0.080} = 33 \text{ A.}$$

Power dissipated ($I^2 R$, with $I^2 = 1056 \text{ A}^2$):

- Good battery's internal resistance: $1056 \times 0.020 = 21 \text{ W}$.
- Cables: $1056 \times 0.010 = 11 \text{ W}$.
- Flat battery's internal resistance: $1056 \times 0.050 = 53 \text{ W}$.

The flat battery also *absorbs* $\mathcal{E}I = 10.0 \times 32.5 = 325 \text{ W}$ as stored chemical energy. That is the useful part of the charging.

P17.10

$$\tau = RC = (2000)(1.0 \times 10^{-3}) = 2.0 \text{ s}, \quad t_{90\%} = \tau \ln 10 = 2.0(2.30) = 4.6 \text{ s.}$$

This is why a camera flash needs a few seconds to “recharge” between shots.

P17.11

(a) $I = 230/(5.0 \times 10^4) = 4.6 \text{ mA}$. This is clearly felt and painful, but generally below the “let-go” threshold.

(b) $I = 230/(1.0 \times 10^3) = 230 \text{ mA}$. This is in the range that can cause **ventricular fibrillation**, which is potentially fatal.

Catheter patients. Normally, most of the body's resistance is in the skin, and current entering through the skin spreads out over the whole chest, so only a small fraction passes through the heart. A catheter or pacing wire inside the heart delivers current *directly* to the heart muscle. It bypasses the skin's resistance and concentrates the current in a tiny region. Currents of 10–100 μA , far too small to feel, can then disturb the heart's rhythm. This is called *microshock*. Equipment connected to such patients must meet very strict limits on leakage current, typically 10 μA .

P17.12

Output of the bridge. Each side of the bridge is a voltage divider. One side has two fixed resistors R , so its midpoint sits at $V/2$. The other side has the gauge $R + \Delta R$ and a fixed R , so its midpoint sits at $V(R + \Delta R)/(2R + \Delta R)$. The output is the difference:

$$V_{\text{out}} = V \left[\frac{R + \Delta R}{2R + \Delta R} - \frac{1}{2} \right] = \frac{V \Delta R}{2(2R + \Delta R)} \approx \frac{V \Delta R}{4R} \quad (\Delta R \ll R).$$

Numbers. $\Delta R = 0.0020 \times 350 = 0.70 \, \Omega$:

$$V_{\text{out}} \approx \frac{10(0.70)}{4(350)} = 5.0 \, \text{mV}.$$

This is a small signal, easily amplified, and the bridge cancels the much larger common voltage of 5 V. Using two or four active gauges (“half” or “full” bridges) increases the output and also compensates for temperature changes. Strain gauges like this are used in load cells, weighing scales, structural monitoring, and force plates for gait analysis.

P17.13

(a) **Derivation.** Between collisions, each electron accelerates at $a = eE/m$. On average it has been accelerating for a time τ since its last collision, which randomised its direction. So its average drift velocity is

$$v_d = \frac{eE\tau}{m}.$$

Then

$$J = nev_d = \frac{ne^2\tau}{m}E = \sigma E \Rightarrow \rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}.$$

(b) **Collision time for copper:**

$$\tau = \frac{m}{ne^2\rho} = \frac{9.11 \times 10^{-31}}{(8.5 \times 10^{28})(1.602 \times 10^{-19})^2(1.7 \times 10^{-8})} = \frac{9.11 \times 10^{-31}}{3.7 \times 10^{-17}} = 2.5 \times 10^{-14} \, \text{s}.$$

(c) **Mean free path:**

$$\lambda = v_F\tau = (1.6 \times 10^6)(2.5 \times 10^{-14}) = 3.9 \times 10^{-8} \, \text{m} = 39 \, \text{nm}.$$

That is about 150 atomic spacings.

What the electrons collide with. They clearly do not bounce off every ion they pass. Quantum mechanics explains why: an electron wave travels through a *perfect* lattice without being scattered at all. Resistance comes only from departures from perfect order, namely thermal vibrations of the lattice (phonons), impurities and defects. That is why resistance falls as metals are cooled and purified.

P17.14

(a) Maximum power.

$$P = \frac{\mathcal{E}^2 R}{(R + r)^2}, \quad \frac{dP}{dR} = \mathcal{E}^2 \frac{(R + r)^2 - 2R(R + r)}{(R + r)^4} = \mathcal{E}^2 \frac{r - R}{(R + r)^3}.$$

This is zero when $R = r$. Then $P_{\max} = \mathcal{E}^2/4r$.

(b) Efficiency.

$$\eta = \frac{I^2 R}{I^2 (R + r)} = \frac{R}{R + r} = \frac{1}{2} \quad \text{at } R = r.$$

(c) **Different goals.** A power grid exists to deliver energy *efficiently*. Wasting half of it inside generators and lines would be absurd. So the load's resistance is made much larger than the source's ($R \gg r$), giving efficiencies close to 100%. Where the source is weak and its output must be maximised, such as an antenna feeding a receiver or an amplifier driving a loudspeaker, impedances are deliberately **matched** to extract the greatest signal power and to avoid reflections, even at 50% efficiency.

P17.15

Self-similarity. Removing the first series resistor and the first rung leaves another infinite ladder, which has the same resistance R_{eq} . So the whole ladder is equivalent to R in series with (R in parallel with R_{eq}):

$$R_{\text{eq}} = R + \frac{R R_{\text{eq}}}{R + R_{\text{eq}}}.$$

Multiplying out:

$$R_{\text{eq}}^2 - R R_{\text{eq}} - R^2 = 0 \Rightarrow R_{\text{eq}} = \frac{R(1 + \sqrt{5})}{2} \approx 1.618R,$$

taking the positive root. This is the golden ratio. Ladder networks model transmission lines, and the passive spread of signals along nerve fibres.

P17.16

(a) **Energy in the resistor.** During charging, $i(t) = (\mathcal{E}/R)e^{-t/RC}$, so

$$E_R = \int_0^\infty i^2 R dt = \frac{\mathcal{E}^2}{R} \int_0^\infty e^{-2t/RC} dt = \frac{\mathcal{E}^2}{R} \cdot \frac{RC}{2} = \frac{1}{2} C \mathcal{E}^2.$$

R cancels: a smaller resistance gives a larger but shorter current, and the total heat is the same.

(b) **Energy from the battery.** The battery moves the total charge $Q = C\mathcal{E}$ through its EMF:

$$E_{\text{batt}} = Q\mathcal{E} = C\mathcal{E}^2.$$

The capacitor stores $\frac{1}{2} C \mathcal{E}^2$, which is **exactly half**. The other half is always lost as heat in charging from a fixed voltage source. Charging gradually, by ramping up the source voltage, reduces this loss, which is the idea behind energy-recovering “adiabatic” circuits.

P17.17

(a) **Time constant.** A patch of area A has resistance $R = R_m/A$ (a larger area gives more parallel paths) and capacitance $C = c_m A$. So

$$\tau = RC = \frac{R_m}{A} \cdot c_m A = R_m c_m = (1.0)(0.010) = 10 \text{ ms},$$

independent of the area.

(b) **Decay from 10 mV to 1 mV:**

$$t = \tau \ln 10 = 10 \text{ ms} \times 2.30 = 23 \text{ ms}.$$

(c) **Effect of myelin.** $\tau = R_m c_m$ becomes $(100R_m)(c_m/100) = R_m c_m$, so it is **unchanged**. Myelin speeds conduction in other ways:

- The much smaller capacitance means far less charge is needed to change the voltage of each stretch of axon.
- The much larger membrane resistance means less current leaks out through the walls.

Together, these let a depolarisation spread passively much farther and faster along the inside of the axon. The action potential then needs to be regenerated only at the gaps in the myelin, the nodes of Ranvier, and the impulse “jumps” from node to node (saltatory conduction). This increases conduction speed from about 1 m/s to as much as 100 m/s. In multiple sclerosis, myelin is lost and conduction slows or fails.

P17.18

(a) **Volume from resistance.** For a uniform cylinder, $R = \rho H/A$ and $V = AH$. Eliminating $A = V/H$:

$$R = \frac{\rho H^2}{V} \Rightarrow V = \frac{\rho H^2}{R}.$$

(Clinical devices use the “impedance index” H^2/R , calibrated against reference methods, for exactly this reason.)

(b) **Total body water:**

$$V = \frac{6.5(1.75)^2}{500} = 0.040 \text{ m}^3 = 40 \text{ L},$$

which is about 40 kg of water.

(c) **Composition.**

$$\text{fat-free mass} = \frac{40 \text{ kg}}{0.73} = 54.5 \text{ kg}, \quad \text{fat} = 70 - 54.5 = 15.5 \text{ kg} \approx 22\%.$$

Dehydration. Losing water reduces the conducting volume, so the measured resistance rises. The device then *underestimates* the fat-free mass and *overestimates* the body fat, even though the person’s tissues have not changed. Alcohol, exercise, meals and skin temperature also shift readings. That is why measurements should be taken under standard conditions.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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